

A2: TEC Heating and Cooling Analysis

Assessment code: A2

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Repository: <https://github.com/kapilnaidu1/phys39-naidu-cohen>, branch main

Part 4: Combined graph and fits

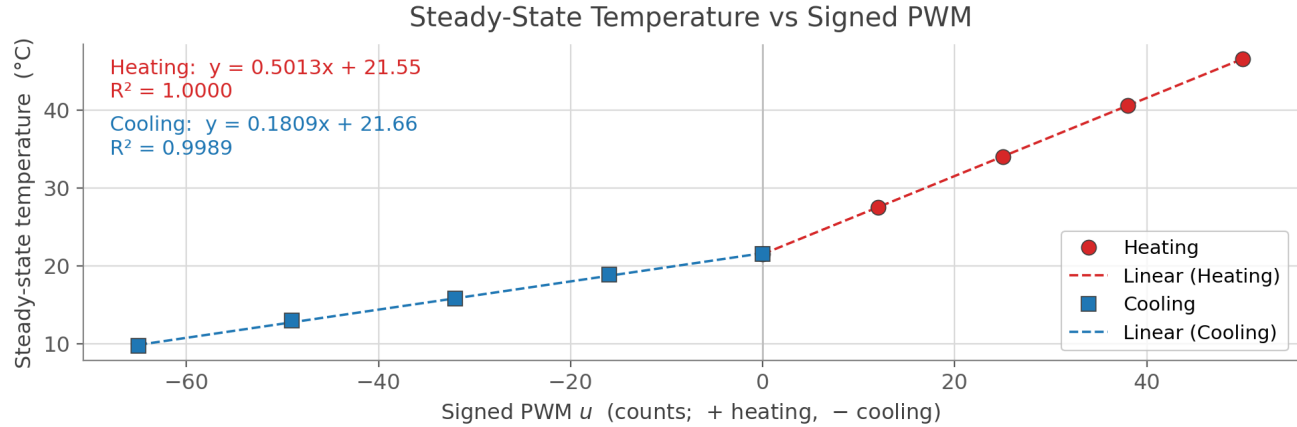


Figure 1. Steady-state plate temperature against the signed PWM command u , positive for heating and negative for cooling, with both branches on one graph. Ten points, five per branch, about the zero-command temperature $T_0 = 21.54$ °C. Each trendline is drawn only over the PWM range used for its own fit: $u = 0$ to $+50$ counts for heating, $u = -65$ to 0 counts for cooling. **Steady-state criterion:** after each step the trace was watched for about three driven time constants ($\tau \approx 70$ s) plus one further minute, and the point was accepted once its net drift over that minute was no larger than the measured noise of 0.16 °C peak to peak. Points that had not fully arrived are reported as the asymptote of an exponential fit, and data/module_04/steady_state.csv records which.

Part 5.1: Measured slopes

Branch	Susceptibility dT/du	Fit range (counts)	R^2	Intercept
Heating	$m_h = 0.5013$ °C per PWM count	$u = 0$ to $+50$	1.0000	21.55 °C
Cooling	$m_c = 0.1809$ °C per PWM count	$u = -65$ to 0	0.9989	21.66 °C

Against signed u both slopes are positive, so the cooling entry is already the magnitude $|x_{T,c}|$. The ratio is $r = m_h / m_c = 0.5013 / 0.1809 = 2.771$, dimensionless because both slopes carry the same units.

Curvature and choice of fit range. Neither data set shows visible curvature, so each branch was fitted over its full measured range rather than a restricted sub-range; the ranges above are simply the PWM values used in Part 3. The branches are fitted separately because the slope changes abruptly at $u = 0$, not because either bends. The intercepts differ by 0.11 °C and the cooling one sits 0.12 °C above T_0 , consistent with the room drifting over a two-hour session.

Part 5.2: PWM and the slope-ratio model

Step 1. The steady-state balance

With C the heat capacity of the object and G the effective passive conductance to the room, $C dT/dt = Q_{TEC} - G(T - T_0)$. At steady state $dT/dt = 0$, so the individual flows are not zero but their sum is, and $G(T - T_0) = Q_{TEC}$. Taking the two terms in turn: **heating**, Peltier transport and Joule dissipation both carry heat *into* the object while conduction carries heat *out of* it to the cooler room; **cooling**, Peltier transport carries heat *out of* the object, Joule dissipation still carries heat *into* it, and conduction carries heat *into* it from the warmer room.

Step 2. PWM current averages

Over one PWM period τ the bridge delivers the on-state current I for a time $D\tau$ and zero for the remaining $(1-D)\tau$, with duty $D = |u| / 255$. Both integrands vanish outside the on-interval, so each integral is just the integrand times $D\tau$:

$$\begin{aligned} \langle I \rangle &= (1/\tau) \int_0^\tau I(t) dt = (1/\tau) \cdot I \cdot D\tau = D I \\ \langle I^2 \rangle &= (1/\tau) \int_0^\tau I^2(t) dt = (1/\tau) \cdot I^2 \cdot D\tau = D I^2 \end{aligned}$$

$\langle I^2 \rangle$ is not $\langle I \rangle^2 = D^2 I^2$: squaring and averaging do not commute, and for a two-level waveform the two differ by a factor $1/D$. Peltier transport follows $\langle I \rangle$ and Joule heating follows $\langle I^2 \rangle$, and under PWM both are linear in D , so the susceptibility is constant and the branches are straight. Had the Joule term gone as D^2 the susceptibility would vary with duty and the branches would curve measurably; the measured R^2 of 1.0000 and 0.9989 says they do not. With a DAC supplying a steady current of variable amplitude instead, $\langle I^2 \rangle = \langle I \rangle^2$ would hold, so the distinction is a property of PWM.

Step 3. The slope ratio

With signed duty $d = u / 255$ and $D = |d|$, Peltier transport reverses with the current and Joule heating does not, so $Q_{\text{TEC}} = d Q_p + |d| Q_j$. Substituting each branch into $G(T - T_0) = Q_{\text{TEC}}$ and differentiating with respect to d :

$$\begin{aligned} \text{heating, } d > 0: \quad G(T_h - T_0) &= d(Q_p + Q_j) \rightarrow T_h - T_0 = d(Q_p + Q_j)/G \rightarrow dT_h/dd = (Q_p + Q_j)/G \\ \text{cooling, } d < 0: \quad G(T_c - T_0) &= d(Q_p - Q_j) \rightarrow T_c - T_0 = d(Q_p - Q_j)/G \rightarrow dT_c/dd = (Q_p - Q_j)/G \end{aligned}$$

Both are positive while $Q_p > Q_j$, and the heating one is larger. Since $d = u / 255$, each measured slope carries the same factor $1/255$, and that factor and G are common to both branches, so both cancel from the ratio:

$$\begin{aligned} r = m_h/m_c &= (Q_p + Q_j) / (Q_p - Q_j) \Rightarrow Q_j/Q_p = (r - 1)/(r + 1) \quad \text{check: } r = 2 \text{ gives } 1/3 \text{ exactly} \\ r = 2.771: \quad Q_j/Q_p &= (2.771 - 1) / (2.771 + 1) = 1.771 / 3.771 = 0.470 \end{aligned}$$

Joule heat reaching the object face is 47.0% of the Peltier pumping. Both are heat rates in watts, so the ratio is dimensionless. G cancels from r , which matters because G was never measured.

Part 5.3: Laird data-sheet calculation

Source. Laird CP14-127-045-L2-W4.5, MFG part 58910-501, from the SPECIFICATIONS table on the first page of the CP14-127-045 data sheet, hot-side temperature $T_h = 27.0^\circ\text{C}$ column.

Quantity	Value	Meaning, and the operating condition attached
R_M	1.50 Ω	Ohmic resistance of the 127 couples in series, quoted at $T_h = 27^\circ\text{C}$ and rising to 1.68 Ω at 50°C
I_{max}	8.6 A	The current giving the largest temperature difference, not a safety ceiling. Condition: I at ΔT_{max}
$Q_{c,\text{max}}$	71.3 W	Most heat the cold face can absorb. Condition: $\Delta T = 0$, no difference to pump against
ΔT_{max}	70.5 $^\circ\text{C}$	Largest sustainable temperature difference. Condition: $Q_c = 0$, no heat load

The last two sit at opposite ends of one trade-off: maximum pumping needs zero temperature difference, maximum difference needs zero load. At the data-sheet condition $\Delta T = 0$, so conduction through the module vanishes and the symmetric model puts half the Joule heat on each face:

$$\begin{aligned} Q_{j,\text{max}} &= \frac{1}{2} I_{\text{max}}^2 R_M = \frac{1}{2} (8.6 \text{ A})^2 (1.50 \Omega) = \frac{1}{2} (73.96 \text{ A}^2)(1.50 \Omega) = 55.47 \text{ W} \\ Q_{c,\text{max}} &= Q_{p,\text{max}} - Q_{j,\text{max}} \rightarrow Q_{p,\text{max}} = 71.3 \text{ W} + 55.47 \text{ W} = 126.77 \text{ W} \\ r_{\text{Laird,max}} &= (Q_{p,\text{max}} + Q_{j,\text{max}}) / (Q_{p,\text{max}} - Q_{j,\text{max}}) = (126.77 + 55.47) / (126.77 - 55.47) = 182.24 / 71.30 = 2.556 \end{aligned}$$

Every quantity above is a heat rate in watts, so $r_{\text{Laird,max}}$ is dimensionless and directly comparable with the measured r .

Cross-check: $V_{\text{max}}/I_{\text{max}} = 13.9 \text{ V} / 8.6 \text{ A} = 1.616 \Omega$, 8% above R_M , because at ΔT_{max} the module develops a Seebeck back-EMF: $V_{\text{max}} = I_{\text{max}} R_M + S \Delta T_{\text{max}}$ gives $S = 0.0142 \text{ V/K}$, reasonable for 127 bismuth telluride couples. The right rows were read.

Part 5.4: Compare the ratios

	r	Q_j/Q_p	Condition
Measured, this apparatus	2.771	0.470	PWM drive, finite ΔT , current set by the supply and bridge
Laird, at I_{max}	2.556	0.438	steady DC at 8.6 A, $\Delta T = 0$
ratio	1.084	1.073	agreement to 8%

The two agree to 8%, closer than the comparison warrants, since the data-sheet figures describe a condition this apparatus never reaches:

- **D = 1 does not mean $I = I_{\text{max}}$.** Full duty only means the bridge is continuously on. The current is set by the 12 V supply and its limit, the H-bridge forward drop, wiring resistance and R_M , which together hold the bench current well below 8.6 A.
- **PWM, not steady DC.** The bench current is chopped; the data sheet assumes a steady one.
- **Finite ΔT and passive heat paths.** The data-sheet relation is stated at $\Delta T = 0$, while the plate ran 25°C above and 12°C below ambient. There the module's own conduction and the mounting and air paths are not negligible, and all of them are absorbed into the single G .
- **Coefficients move with temperature, and one slope is fitted per branch.** R_M rises from 1.50 to 1.68 Ω between 27 and 50°C , so a single fit averages over a range in which the coefficients are not strictly constant.

Direction of the difference. The measured ratio is the larger, yet running below I_{max} should *reduce* Q_j/Q_p , since that ratio scales as I . The opposite sign points to dissipation this model assigns to the object face that the data sheet does not carry, such as lead and contact resistance in series with the module.

Part 5.4: Passive conduction

When the object is hotter than the room, passive heat flows **out of** the object. When it is colder, passive heat flows **into** it. Either way the flow drives the object back toward room temperature, so conduction opposes the TEC on both branches, which is why it enters as $-G(T - T_0)$ with a single positive G .

It cannot by itself explain the unequal slope magnitudes. The same G appears in the denominator of both dT_h/dd and dT_c/dd , so it sets how steep the branches are, a larger G flattening both, but it cancels exactly from their ratio: a symmetric term cannot produce an asymmetric result. The asymmetry survives only in the numerators, where Q_j is added on the heating branch and subtracted on the cooling one, because Peltier transport reverses with the current and Joule heating does not.